



ITEC624 - Network Security & Applications

Semester 2 - 2025

Assessment 3

Skill-Based Tasks & Case Study

**Submitted By
Amir Tamang**

**Student ID
S00393664**

Word Count: 2229

Feed Forward:

Reflection Questions	Feedback and Improvement Actions
1. In your opinion, how do you think previous project inputs contributed to the development of this report?	As stated in previous reviews, this needs a more thorough explanation of the technical decisions taken, which I have tried to cover in this report: for example, the considerations made against the costs, performance, and the meeting of objectives set by ABC Company in choosing Azure PaaS and parts of the hardware.
2. For future projects, what type of feedback would you find most helpful?	Detailed feedback is what helps me tune-especially those areas that need work, like technical detail, clarity, or structure. When I know exactly where I need to add evidence or revise my arguments, I can construct stronger, more targeted arguments in future assignments.
3. Were you having problems trying to understand the use of comments from previous assignments?	Some of the previous critiques were too general to locate specific areas that needed attention. For instance, comments that highlight specific passages or ideas that require more careful consideration or evidence are of more value than the general advice to "improve the analysis."
4. How has your strategy changed considering earlier criticism?	Previous remarks emphasized that comments should be short and fact-based. Therefore, in this research work, I tried to give the shortest explanations and examples that will ensure that things are clear and support my recommendations effectively, such as comprehensive estimates of costs for cloud services.
5. About anything, how do you think the feedback process could be improved to help you learn more?	This would also be assisted if recommendations were added concerning readings or resources on hard subjects, such as cloud architecture models. Additionally, apart from this, feedback should include a summary of strengths and areas of development. That would really help me understand and lift the standard of my work in general.

Checklist:

My submitted assignment report is within the specified word limit.	✓
I have included references using the specified referencing style.	✓
I have correctly cited all my sources and references.	✓
I have formatted my report as per the specifications.	✓
I have checked my Turnitin report to ensure the similarity report is within the acceptable level(below 20% similarity)	✓
I have actioned the feedback advice provided to me from the previous assignment feedback.	✓
I have completed proofreading and checked for spelling and grammar.	✓
I have submitted my work before the due date/time.	✓
I have submitted feed feed-forward template along with my assignment submission.	✓

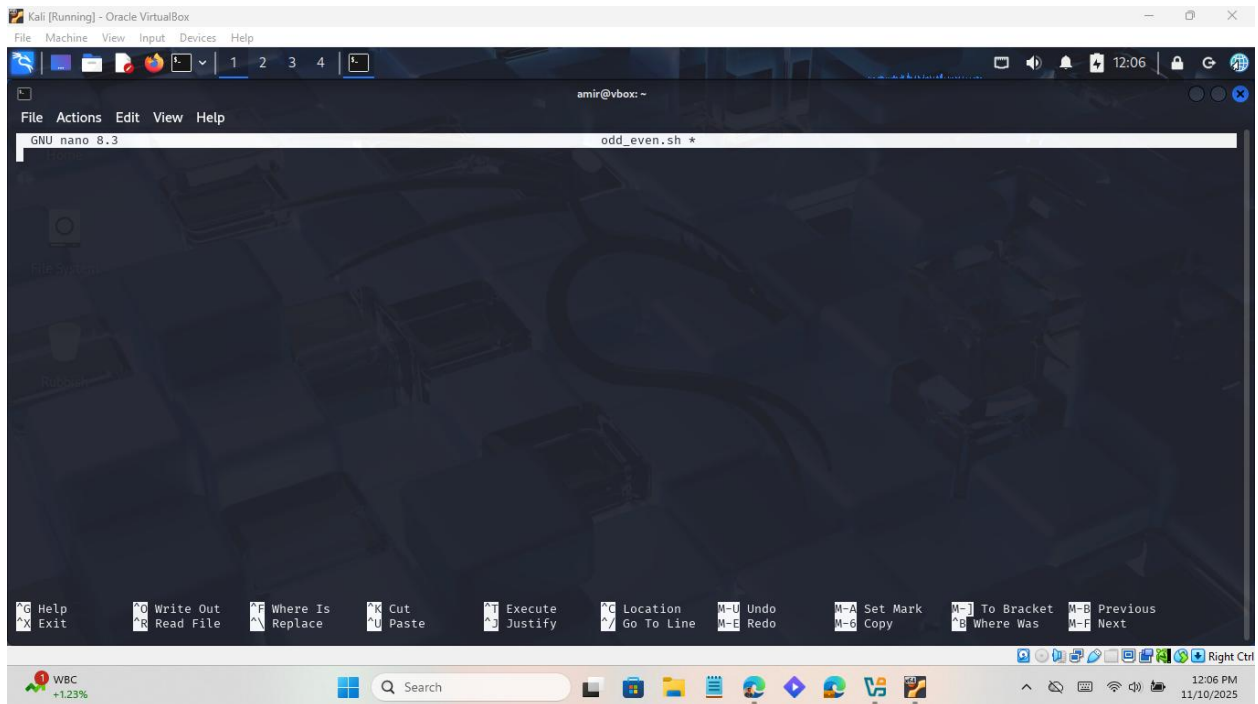
Table of Contents

Part 1: Skill-based assessment	1
Q1. Shell script: odd/even with n repetitions.....	1
Q2. Advanced directory permissions and ACLs	3
Q3. Password cracking.....	12
Q4. Nmap reconnaissance and NSE vulnerability assessment	18
Part 2: Case Study.....	Error! Bookmark not defined.
Abstract	Error! Bookmark not defined.
Introduction.....	Error! Bookmark not defined.
Understanding and Managing Risks	Error! Bookmark not defined.
Protecting Business Resources	Error! Bookmark not defined.
Creating a Safe and Secure Work Environment.....	Error! Bookmark not defined.
Conclusion	Error! Bookmark not defined.
References.....	Error! Bookmark not defined.

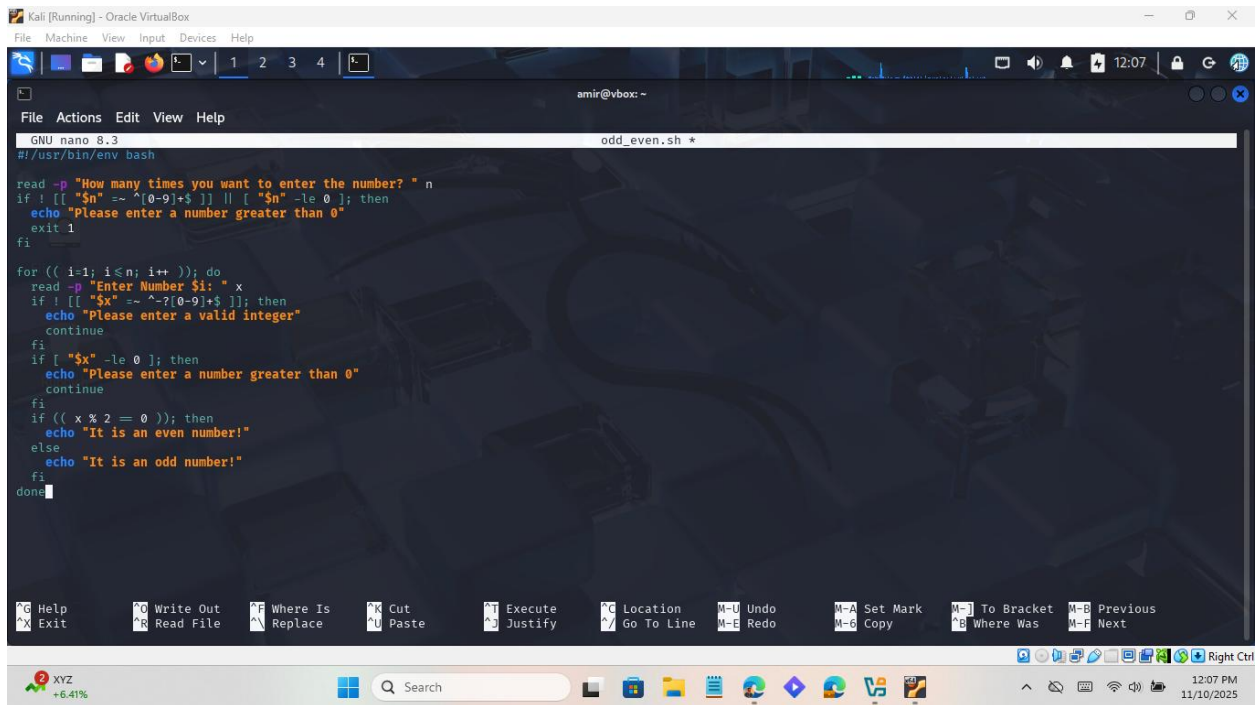
Part 1: Skill-based assessment

Q1. Shell script: odd/even with n repetitions

1.1. Create the script (odd_even.sh)



1.2. Code for odd_even.sh file:

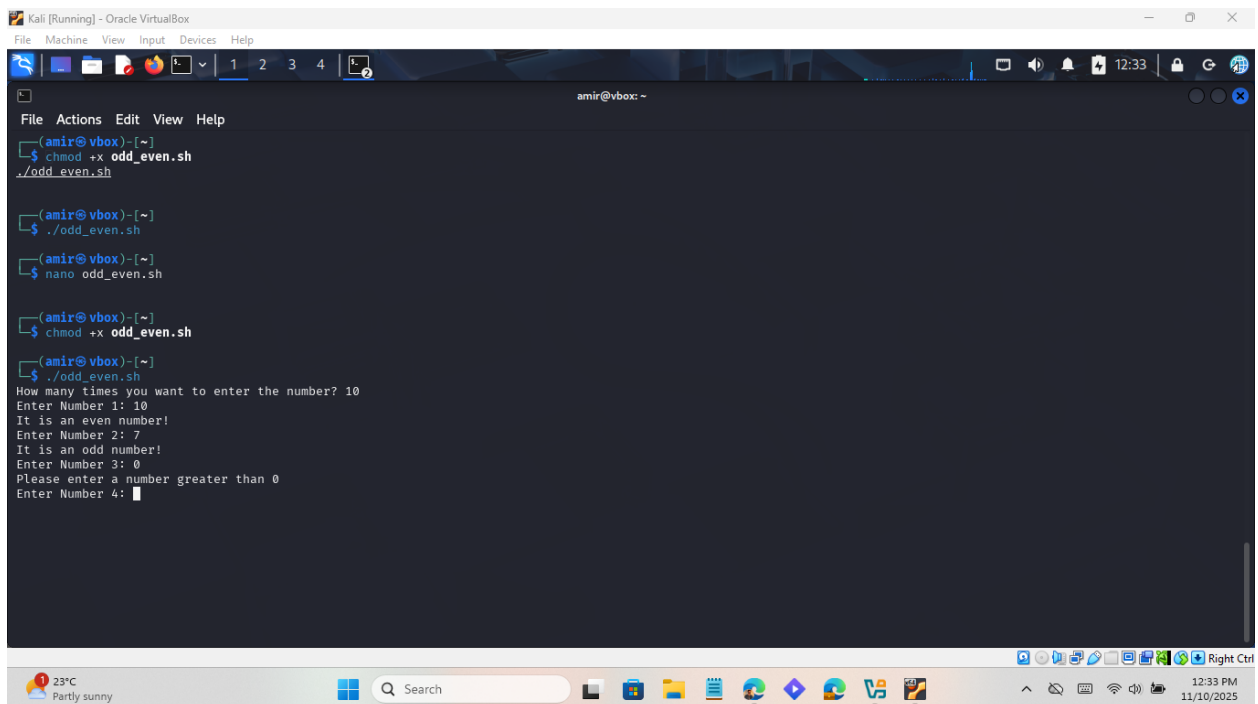


```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
amir@vbox: ~
GNU nano 8.3 odd_even.sh
#!/usr/bin/env bash

read -p "How many times you want to enter the number? " n
if ! [[ "$n" =~ ^[0-9]+$ ]] || [ "$n" -le 0 ]; then
    echo "Please enter a number greater than 0"
    exit 1
fi

for (( i=1; i<=n; i++ )); do
    read -p "Enter Number $i: " x
    if ! [[ "$x" =~ ^[0-9]+$ ]]; then
        echo "Please enter a valid integer"
        continue
    fi
    if [ "$x" -le 0 ]; then
        echo "Please enter a number greater than 0"
        continue
    fi
    if (( x % 2 == 0 )); then
        echo "It is an even number!"
    else
        echo "It is an odd number!"
    fi
done
```

1.3. Make executable and run



```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
amir@vbox: ~
File Actions Edit View Help
$ chmod +x odd_even.sh
$ ./odd_even.sh

$ nano odd_even.sh

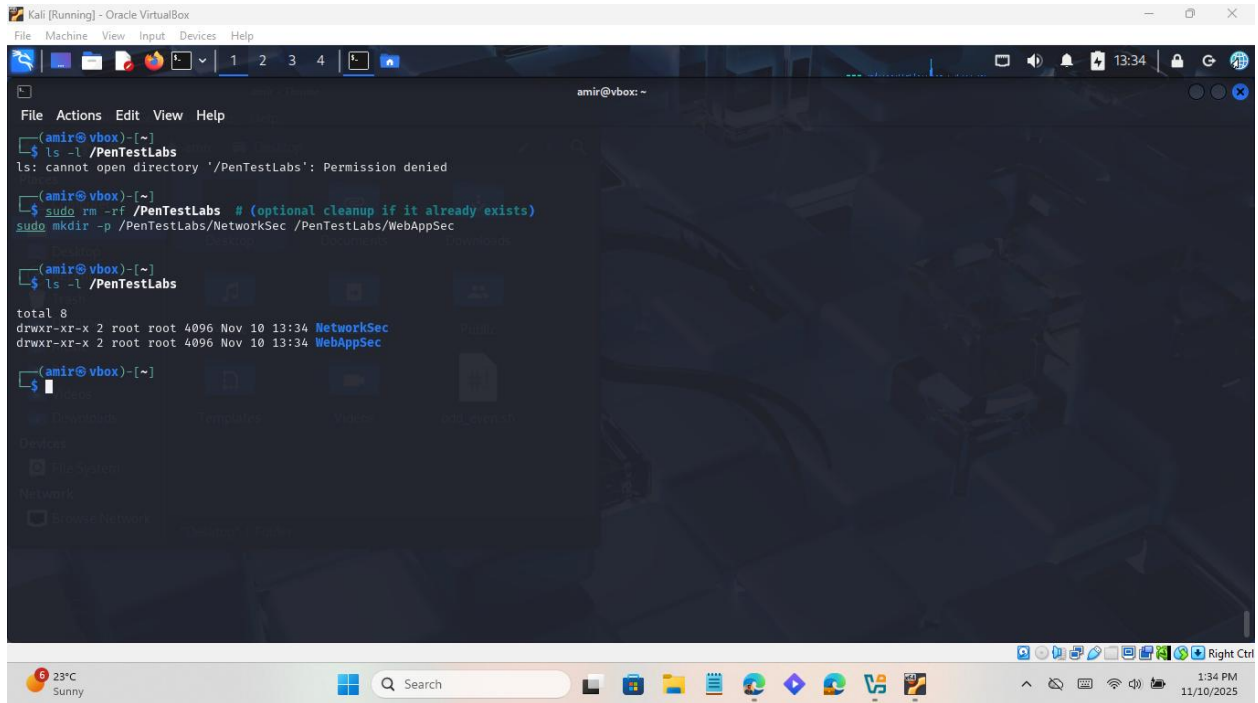
$ chmod +x odd_even.sh

$ ./odd_even.sh
How many times you want to enter the number? 10
Enter Number 1: 10
It is an even number!
Enter Number 2: 7
It is an odd number!
Enter Number 3: 0
Please enter a number greater than 0
Enter Number 4: 
```

Q2. Advanced directory permissions and ACLs

I will create users/groups, set directory ownership/permissions, and verify with ACLs.

2.1. Create the directory structure



The screenshot shows a terminal window titled "Kali [Running] - Oracle VirtualBox". The user is logged in as "amir@vbox: ~". The terminal output is as follows:

```
File Actions Edit View Help
(amir@vbox)-[~]
$ ls -l /PenTestLabs
ls: cannot open directory '/PenTestLabs': Permission denied

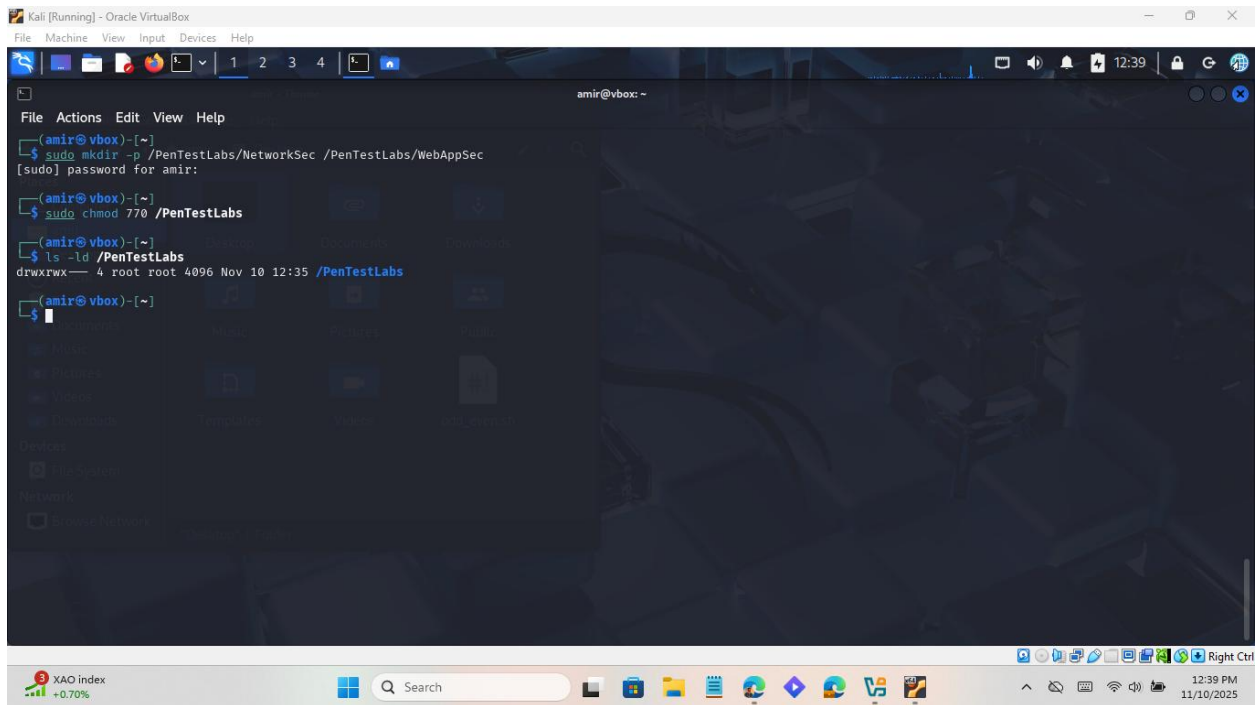
(amir@vbox)-[~]
$ sudo rm -rf /PenTestLabs # (optional cleanup if it already exists)
$ sudo mkdir -p /PenTestLabs/NetworkSec /PenTestLabs/WebAppSec

(amir@vbox)-[~]
$ ls -l /PenTestLabs
total 8
drwxr-xr-x 2 root root 4096 Nov 10 13:34 NetworkSec
drwxr-xr-x 2 root root 4096 Nov 10 13:34 WebAppSec

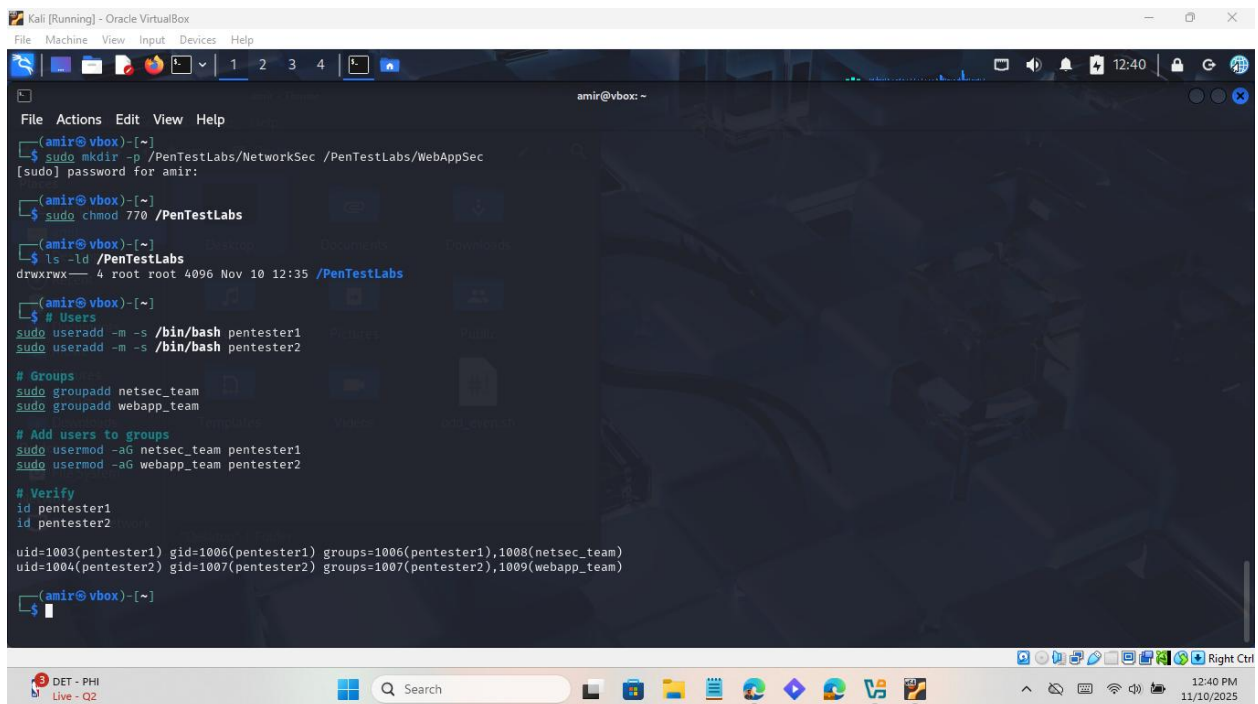
(amir@vbox)-[~]
$
```

The terminal window is running on a Kali Linux virtual machine. The user attempts to list the contents of the /PenTestLabs directory but receives a "Permission denied" error. They then use sudo to remove the directory (commented out) and create the directory structure: /PenTestLabs/NetworkSec and /PenTestLabs/WebAppSec. Finally, they list the contents of /PenTestLabs, showing two directories: NetworkSec and WebAppSec, both with permissions drwxr-xr-x, owned by root, and created on Nov 10 13:34.

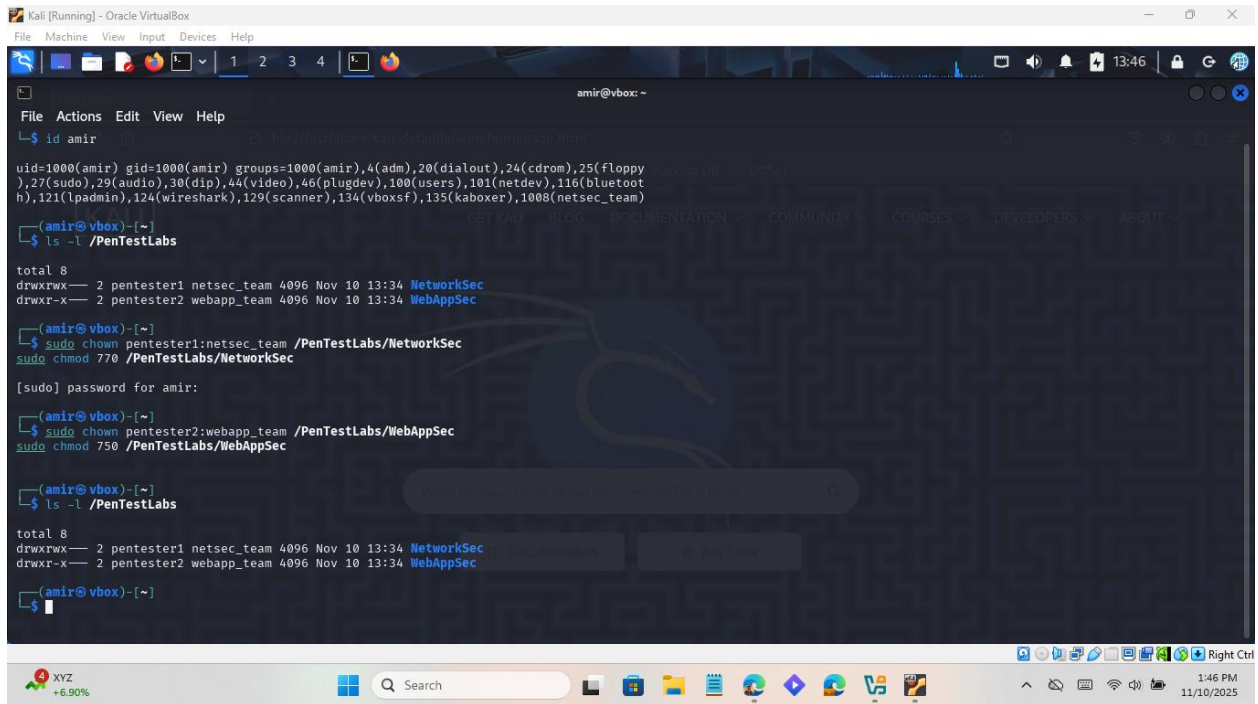
2.2. Set PenTestLabs permissions



2.3. Create users and groups



2.4. Set subdirectory permissions



```
amir@vbox: ~  
File Actions Edit View Help  
└─$ id amir  
uid=1000(amir) gid=1000(amir) groups=1000(amir),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),116(bluetooth),121(lpadmin),124(wireshark),129(scanner),134(vboxsf),135(kaboxer),1008(netsec_team)  
  
amir@vbox: ~  
└─$ ls -l /PenTestLabs  
total 8  
drwxrwx--- 2 pentester1 netsec_team 4096 Nov 10 13:24 NetworkSec  
drwxr-x--- 2 pentester2 webapp_team 4096 Nov 10 13:34 WebAppSec  
  
amir@vbox: ~  
└─$ sudo chown pentester1:netsec_team /PenTestLabs/NetworkSec  
sudo chmod 770 /PenTestLabs/NetworkSec  
[sudo] password for amir:  
amir@vbox: ~  
└─$ sudo chown pentester2:webapp_team /PenTestLabs/WebAppSec  
sudo chmod 750 /PenTestLabs/WebAppSec  
  
amir@vbox: ~  
└─$ ls -l /PenTestLabs  
total 8  
drwxrwx--- 2 pentester1 netsec_team 4096 Nov 10 13:24 NetworkSec  
drwxr-x--- 2 pentester2 webapp_team 4096 Nov 10 13:34 WebAppSec  
amir@vbox: ~  
└─$
```

2.5. Verify with getfacl and user tests

As pentester1

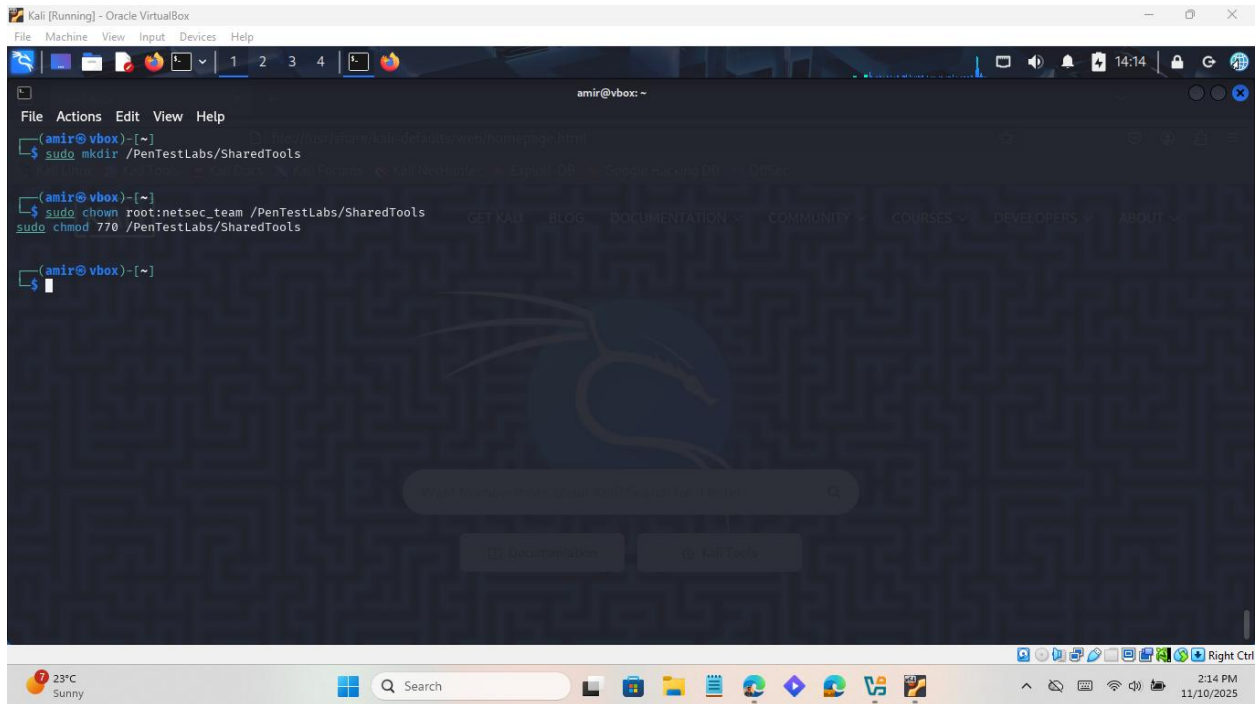
```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
amir@vbox: ~
File Actions Edit View Help
[sudo] password for amir:
$ sudo chown pentester2:webapp_team /PenTestLabs/WebAppSec
$ sudo chmod 750 /PenTestLabs/WebAppSec
(amir@vbox)-[~]
$ ls -l /PenTestLabs
total 8
drwxrwx--- 2 pentester1 netsec_team 4096 Nov 10 13:34 NetworkSec
drwxr-x--- 2 pentester2 webapp_team 4096 Nov 10 13:34 WebAppSec
(amir@vbox)-[~]
$ sudo -u pentester1 bash -c 'touch /PenTestLabs/NetworkSec/test1.txt && echo "pentester1 can write to NetworkSec"'
$ sudo -u pentester1 bash -c 'touch /PenTestLabs/WebAppSec/deny1.txt || echo "pentester1 denied from WebAppSec"'
pentester1 can write to NetworkSec
touch: cannot touch '/PenTestLabs/WebAppSec/deny1.txt': Permission denied
x pentester1 denied from WebAppSec
(amir@vbox)-[~]
$
```

As pentester2

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
amir@vbox: ~
File Actions Edit View Help
(amir@vbox)-[~]
$ sudo -u pentester2 bash -c 'touch /PenTestLabs/WebAppSec/test2.txt && ls -l /PenTestLabs/WebAppSec'
$ sudo -u pentester2 bash -c 'ls /PenTestLabs/NetworkSec && touch /PenTestLabs/NetworkSec/deny2.txt || echo "No access or no write (expected)"'
total 0
-rw-rw-r-- 1 pentester2 pentester2 0 Nov 10 14:05 test2.txt
ls: cannot open directory '/PenTestLabs/NetworkSec': Permission denied
No access or no write (expected)
(amir@vbox)-[~]
$
```

2.6. SharedTools: both can read/execute; only netsec_team can write

2.6.1. Create the directory and set base ownership and permissions

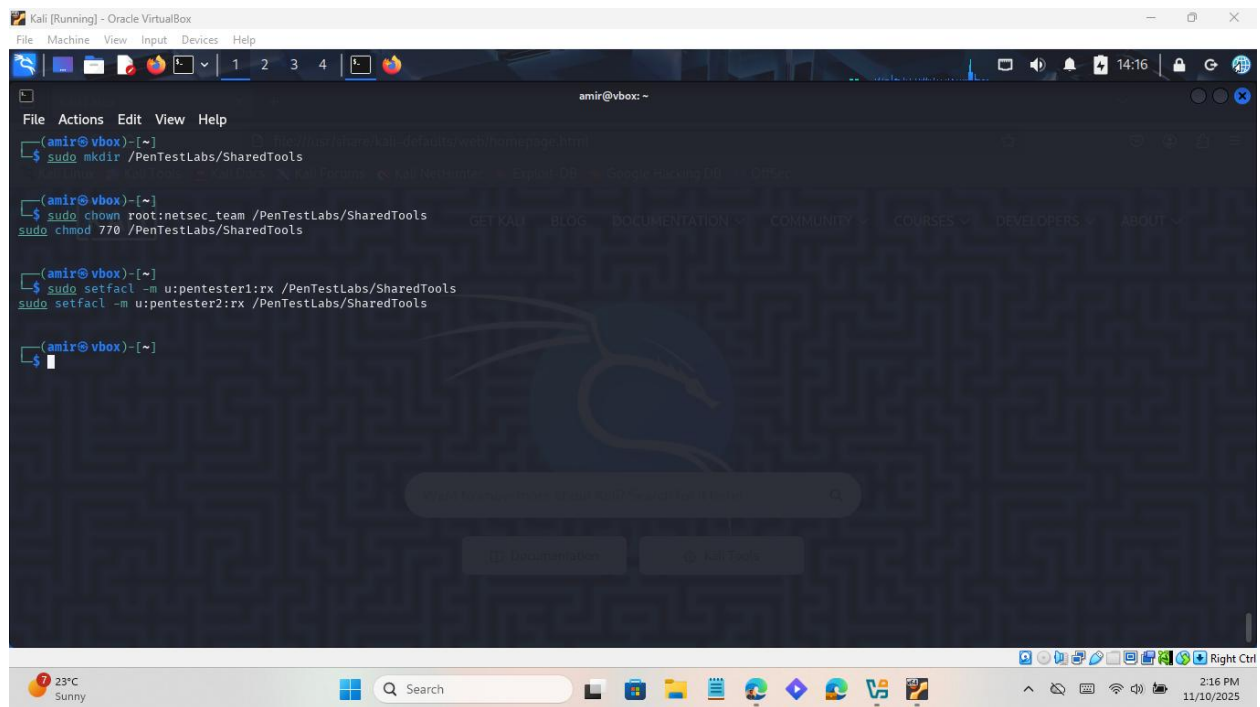


The screenshot shows a Kali Linux terminal window titled "Kali [Running] - Oracle VirtualBox". The terminal displays the following commands and output:

```
amir@vbox: ~  
File Actions Edit View Help  
$ sudo mkdir /PenTestLabs/SharedTools  
$ sudo chown root:netsec_team /PenTestLabs/SharedTools  
$ sudo chmod 770 /PenTestLabs/SharedTools  
$
```

The terminal background features a Kali Linux dragon logo. The window's top bar includes a menu (File, Machine, View, Input, Devices, Help) and a status bar showing the time (14:14) and system icons. The bottom of the image shows a Windows taskbar with various application icons, a search bar, and system information (23°C Sunny, 2:14 PM, 11/10/2025).

2.6.2. Apply ACLs for read/execute access



This allows both users to read and enter the directory but not write.

2.6.3. Verify ACLs

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@vbox: ~
File Actions Edit View Help

(root@vbox)-[~]
# sudo mkdir /PenTestLabs/SharedTools
# sudo chown root:netsec_team /PenTestLabs/SharedTools
# sudo chmod 770 /PenTestLabs/SharedTools
# sudo setfacl -m u:pentester1:rx /PenTestLabs/SharedTools
# sudo setfacl -m u:pentester2:rx /PenTestLabs/SharedTools

mkdir: cannot create directory '/PenTestLabs/SharedTools': File exists

(root@vbox)-[~]
# sudo setfacl -m u:pentester1:rx /PenTestLabs/SharedTools
# sudo setfacl -m u:pentester2:rx /PenTestLabs/SharedTools

(root@vbox)-[~]
# getfacl /PenTestLabs/SharedTools

getfacl: Removing leading '/' from absolute path names
# file: PenTestLabs/SharedTools
# owner: root
# group: netsec_team
user::rwx
user:amir:r-x
user:pentester1:r-x
user:pentester2:r-x
group::rwx
mask::rwx
other::—

(root@vbox)-[~]
```

2.6.4. Access tests

As pentester1 (netsec_team member)

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@vbox: ~
File Actions Edit View Help

(root@vbox)-[~]
# getfacl /PenTestLabs/SharedTools

getfacl: Removing leading '/' from absolute path names
# file: PenTestLabs/SharedTools
# owner: root
# group: netsec_team
user::rwx
user:amir:r-x
user:pentester1:r-x
user:pentester2:r-x
group::rwx
mask::rwx
other::—

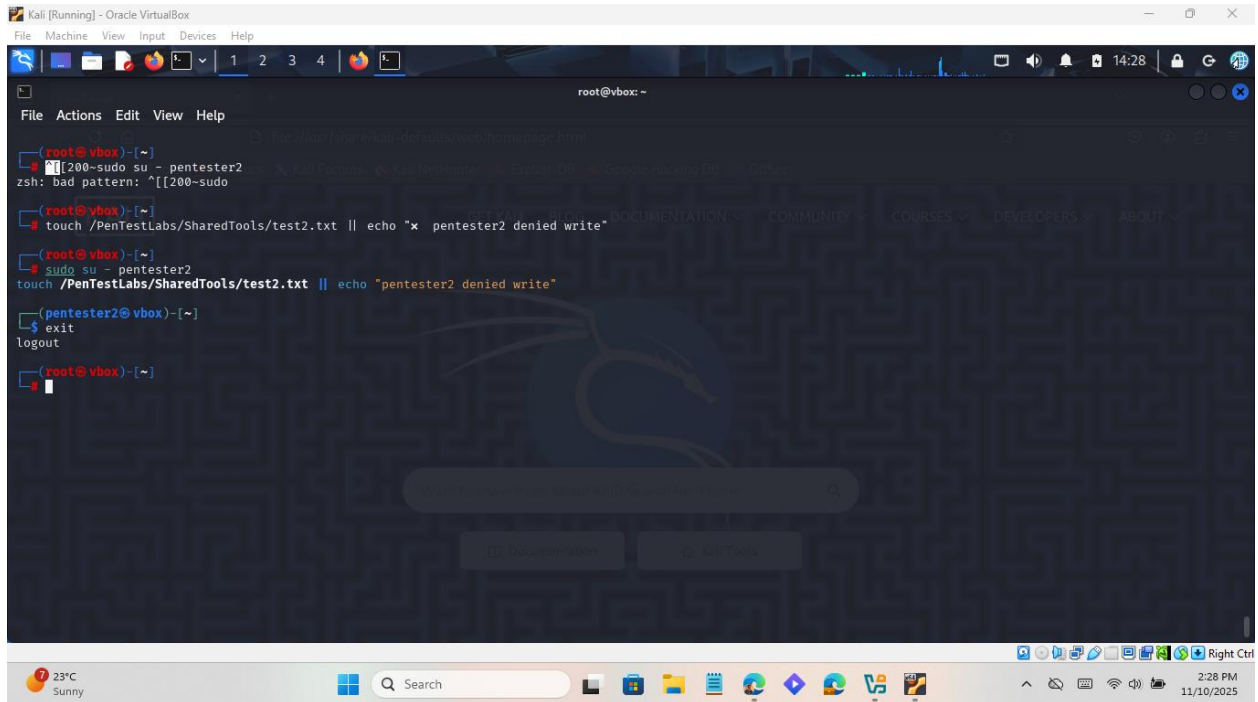
(root@vbox)-[~]
# sudo su - pentester1
touch /PenTestLabs/SharedTools/test1.txt 66 echo "pentester1 can write"

(pentester1@vbox)-[~]
# touch /PenTestLabs/SharedTools/test1.txt 66 echo "pentester1 can write"
touch: cannot touch '/PenTestLabs/SharedTools/test1.txt': Permission denied

(pentester1@vbox)-[~]
# exit
logout
pentester1 can write

(root@vbox)-[~]
```

As pentester2 (not in netsec_team)



```
root@vbox: ~  
File Actions Edit View Help  
[root@vbox]~  
[200-sudo su - pentester2  
zsh: bad pattern: ^[[200-sudo  
[root@vbox]~  
[touch /PenTestLabs/SharedTools/test2.txt || echo "x pentester2 denied write"  
[root@vbox]~  
[sudo su - pentester2  
touch /PenTestLabs/SharedTools/test2.txt || echo "pentester2 denied write"  
[pentester2@vbox]~  
$ exit  
logout  
[root@vbox]~  
[
```

2.7. Supervisor read-only across all

2.7.1. Create a supervisor user


```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@vbox: ~
File Actions Edit View Help

(root@vbox)-[~]
# [[200-sudo su - pentester2
zsh: bad pattern: "[[200-sudo

(root@vbox)-[~]
# touch /PenTestLabs/SharedTools/test2.txt || echo "x pentester2 denied write"
touch /PenTestLabs/SharedTools/test2.txt || echo "pentester2 denied write"

(sudo su - pentester2
touch /PenTestLabs/SharedTools/test2.txt || echo "pentester2 denied write"

(pentester2@vbox)-[~]
$ exit
logout

(root@vbox)-[~]
# sudo useradd supervisor
sudo passwd supervisor

New password:
Retype new password:
passwd: password updated successfully

(root@vbox)-[~]
```

2.7.2. Apply ACL for full access and verify ACL

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@vbox: ~
File Actions Edit View Help

logout

(root@vbox)-[~]
# sudo useradd supervisor
sudo passwd supervisor

New password:
Retype new password:
passwd: password updated successfully

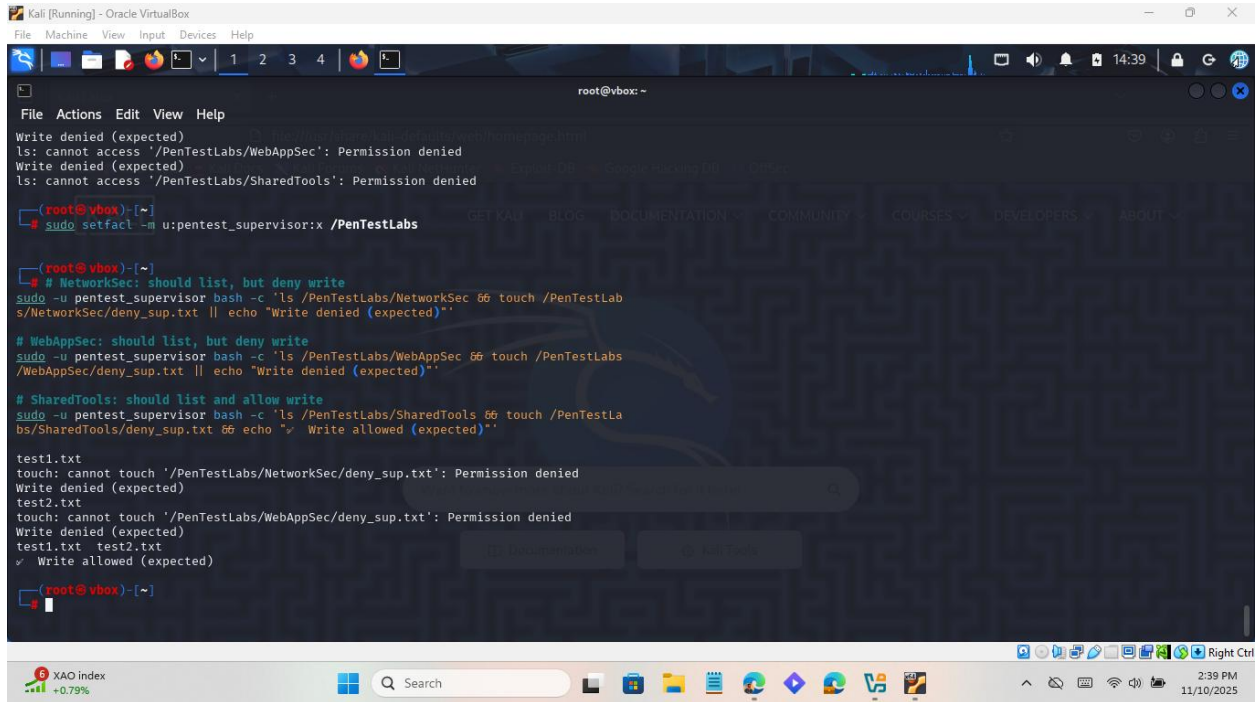
(root@vbox)-[~]
# sudo setfacl -m u:supervisor:rwX /PenTestLabs/SharedTools

(root@vbox)-[~]
# getfacl /PenTestLabs/SharedTools

getfacl: Removing leading '/' from absolute path names
# file: PenTestLabs/SharedTools
# owner: root
# group: netsec_team
user::rwX
user:amir:r-x
user:pentester1:r-x
user:pentester2:r-x
user:supervisor:rwX
group::rwX
mask::rwX
other::—

(root@vbox)-[~]
```

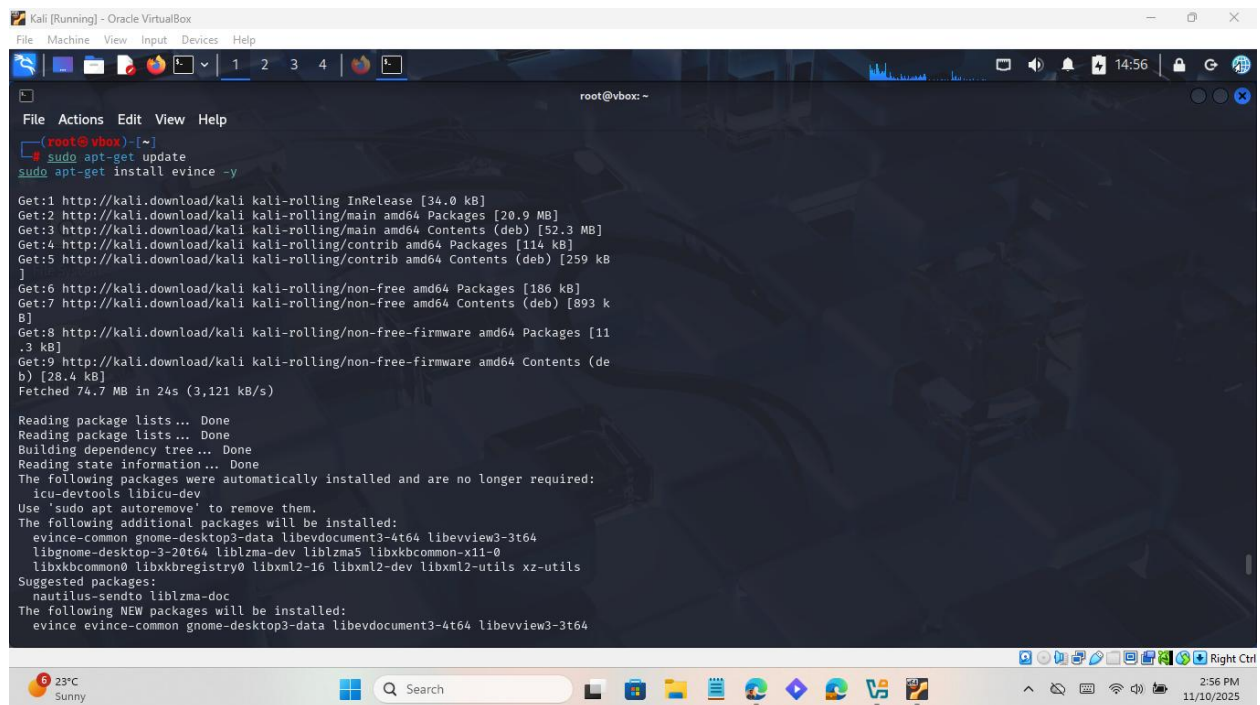
2.7.3. Test



```
root@vbox: ~  
File Actions Edit View Help  
Write denied (expected)  
ls: cannot access '/PenTestLabs/WebAppSec': Permission denied  
Write denied (expected)  
ls: cannot access '/PenTestLabs/SharedTools': Permission denied  
root@vbox: ~  
# sudo setfacl -m u:pentest_supervisor:x /PenTestLabs  
root@vbox: ~  
# # NetworkSec: should list, but deny write  
sudo -u pentest_supervisor bash -c 'ls /PenTestLabs/NetworkSec && touch /PenTestLabs/NetworkSec/deny_sup.txt || echo "Write denied (expected)";  
# WebAppSec: should list, but deny write  
sudo -u pentest_supervisor bash -c 'ls /PenTestLabs/WebAppSec && touch /PenTestLabs/WebAppSec/deny_sup.txt || echo "Write denied (expected)";  
# SharedTools: should list and allow write  
sudo -u pentest_supervisor bash -c 'ls /PenTestLabs/SharedTools && touch /PenTestLabs/SharedTools/deny_sup.txt && echo "Write allowed (expected)";  
test1.txt  
touch: cannot touch '/PenTestLabs/NetworkSec/deny_sup.txt': Permission denied  
Write denied (expected)  
test2.txt  
touch: cannot touch '/PenTestLabs/WebAppSec/deny_sup.txt': Permission denied  
Write denied (expected)  
test1.txt test2.txt  
Write allowed (expected)  
root@vbox: ~  
#
```

Q3. Password cracking

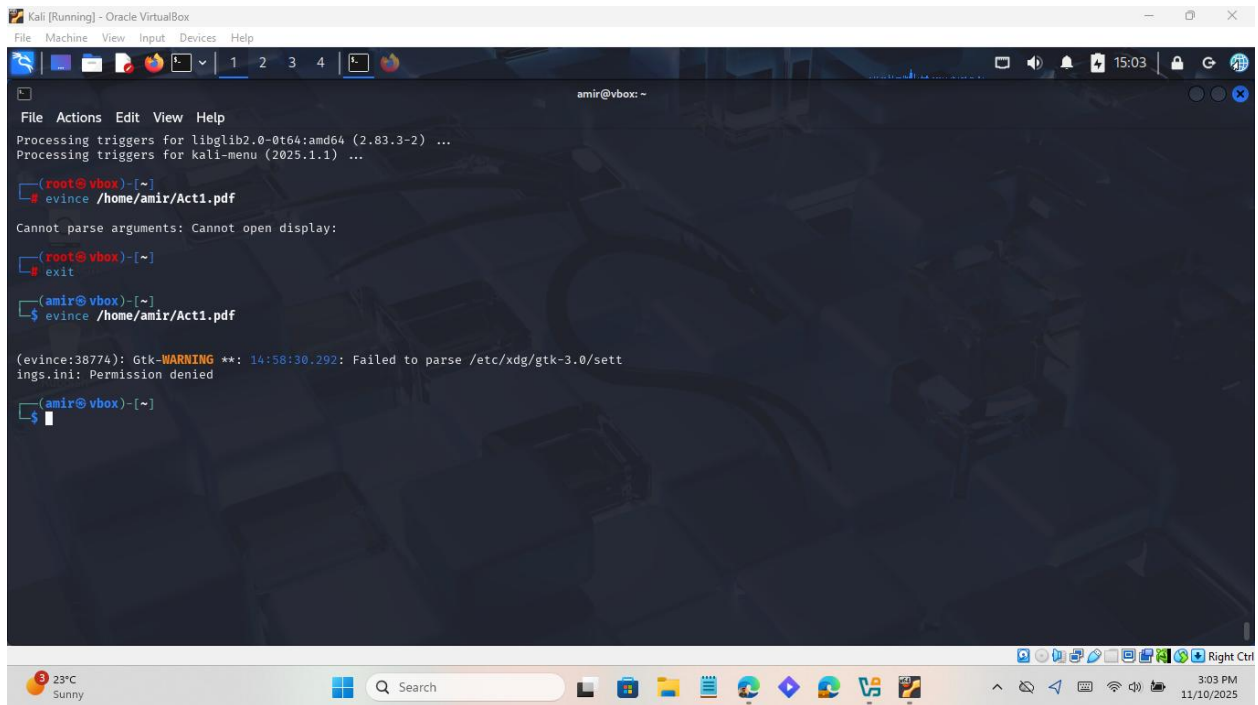
3.1. Install a PDF viewer



```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@vbox: ~
File Actions Edit View Help
root@vbox: ~
$ sudo apt-get update
sudo apt-get install evince -y
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.3 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [259 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [28.4 kB]
Fetched 74.7 MB in 24s (3,121 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
icu-devtools libicu-dev
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
evince-common gnome-desktop3-data libevdocument3-4t64 libevview3-3t64
libgnome-desktop-3-20t64 liblzma-dev liblzma5 libxkbcommon-x11-0
libxkbcommon0 libxkbregistry0 libxml2-16 libxml2-dev libxml2-utils xz-utils
Suggested packages:
nautilus-sendto liblzma-doc
The following NEW packages will be installed:
evince evince-common gnome-desktop3-data libevdocument3-4t64 libevview3-3t64
```

3.2. Act1.pdf: password profiling (social engineering)

3.2.1. Open Act1.pdf file in pdf viewer



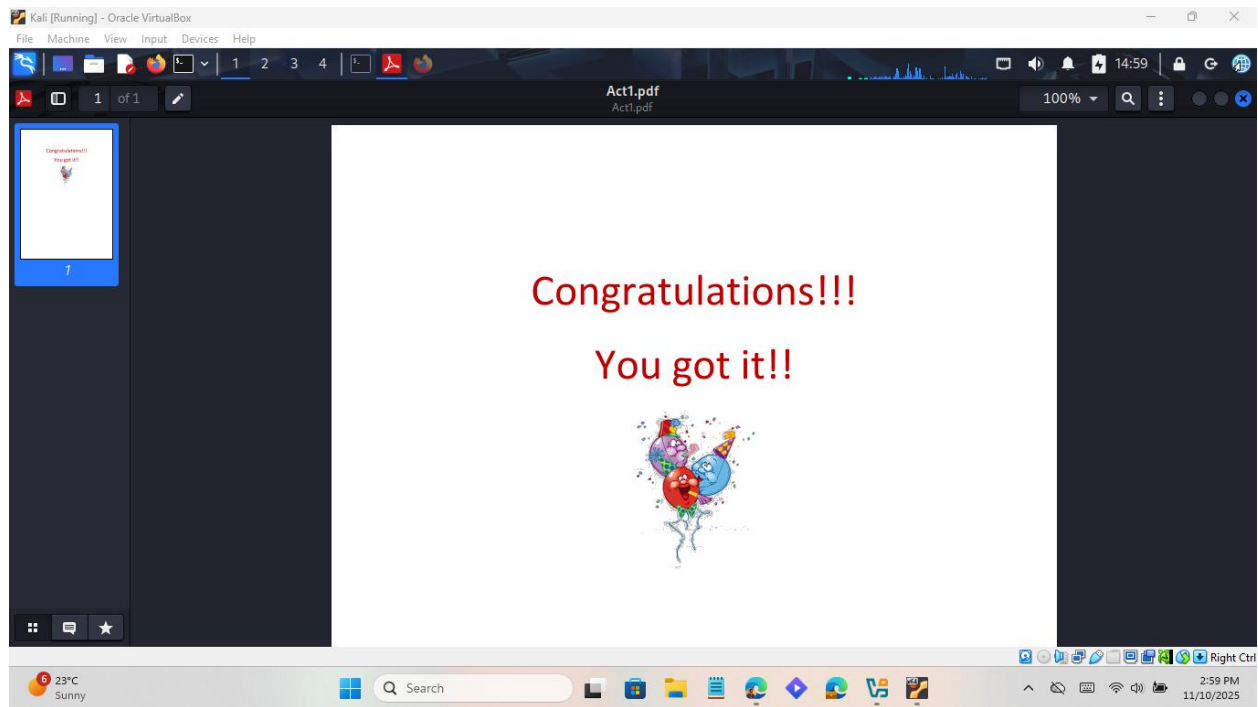
Based on the conversation:

- Jenny + favorite color “purple” + teacher/school cues. Try common human patterns at the PDF prompt:
- purple
- Purple
- jenny
- Jenny
- purplejenny
- jennypurple
- teacher
- english
- school

- purpleglasses

I tried all these password candidates to crack the password. So, the password for the Act1.pdf was **“purple”**. This password unlocked the file.

3.2.2. Unlocked Act1.pdf



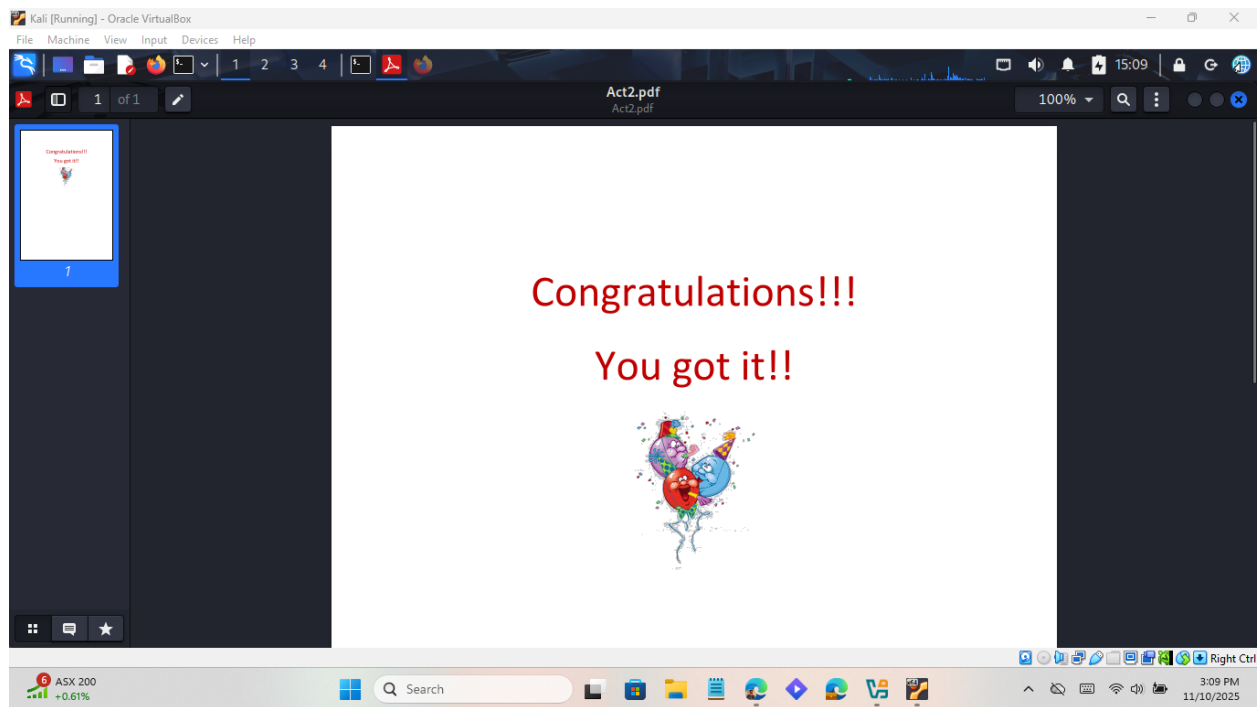
3.3. Act2.pdf: 6-digit numeric brute-force in Kali

3.3.1. Install pdftcrack

3.3.3. Open the PDF with the cracked password

[illegible]

Result:



Q4. Nmap reconnaissance and NSE vulnerability assessment

4.1. Discover active hosts (ping sweep)

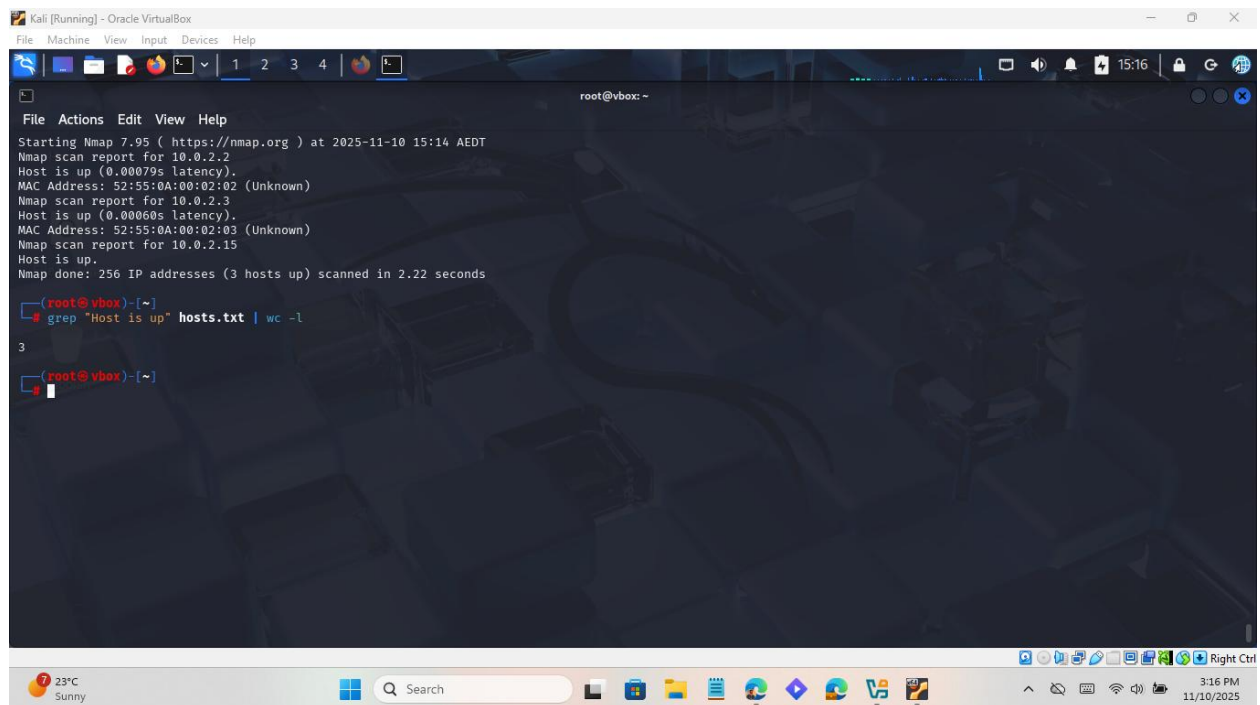
4.1.1. Identify subnet:

```
root@vbox: ~  
File Actions Edit View Help  
(root@vbox)-[~]  
# ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:40:1b:ce brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 81072sec preferred_lft 81072sec  
    inet6 fd17:625c:f037:2:f093:ec98:b6f1:ab4e/64 scope global temporary dynamic  
        valid_lft 85974sec preferred_lft 13974sec  
    inet6 fd17:625c:f037:2:a00:27ff:fe40:1bce/64 scope global dynamic mngtmpaddr noprefixroute  
        valid_lft 85974sec preferred_lft 13974sec  
    inet6 fe80::a00:27ff:fe40:1bce/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(root@vbox)-[~]  
#
```

4.1.2. Run sweep

```
root@vbox: ~  
File Actions Edit View Help  
valid_lft forever preferred_lft forever  
(root@vbox)-[~]  
# sudo nmap -sn 10.0.2.0/24 -oN hosts.txt  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:14 AEDT  
Nmap scan report for 10.0.2.2  
Host is up (0.00079s latency).  
MAC Address: 52:55:0A:00:02:02 (Unknown)  
Nmap scan report for 10.0.2.3  
Host is up (0.00060s latency).  
MAC Address: 52:55:0A:00:02:03 (Unknown)  
Nmap scan report for 10.0.2.15  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.22 seconds  
(root@vbox)-[~]  
#
```


4.1.3. Count active hosts



The screenshot shows a Kali Linux terminal window titled "Kali [Running] - Oracle VirtualBox". The terminal output displays the results of an Nmap scan performed at 2025-11-10 15:14 AEDT. The scan reports three active hosts: 10.0.2.2, 10.0.2.3, and 10.0.2.15. Each host is reported as "Host is up" with a latency of 0.00079s, 0.00060s, and 0.00060s respectively. The MAC address for all three hosts is 52:55:0A:00:02:02 (Unknown). The scan was completed in 2.22 seconds, scanning 256 IP addresses. Below the Nmap output, the user runs the command `grep "Host is up" hosts.txt | wc -l`, which returns the output `3`, indicating that there are three active hosts.

```
File Actions Edit View Help
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:14 AEDT
Nmap scan report for 10.0.2.2
Host is up (0.00079s latency).
MAC Address: 52:55:0A:00:02:02 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.00060s latency).
MAC Address: 52:55:0A:00:02:03 (Unknown)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.22 seconds

(root@vbox)-[~]
# grep "Host is up" hosts.txt | wc -l
3
(root@vbox)-[~]
```

4.1.4. Service version detection on one host

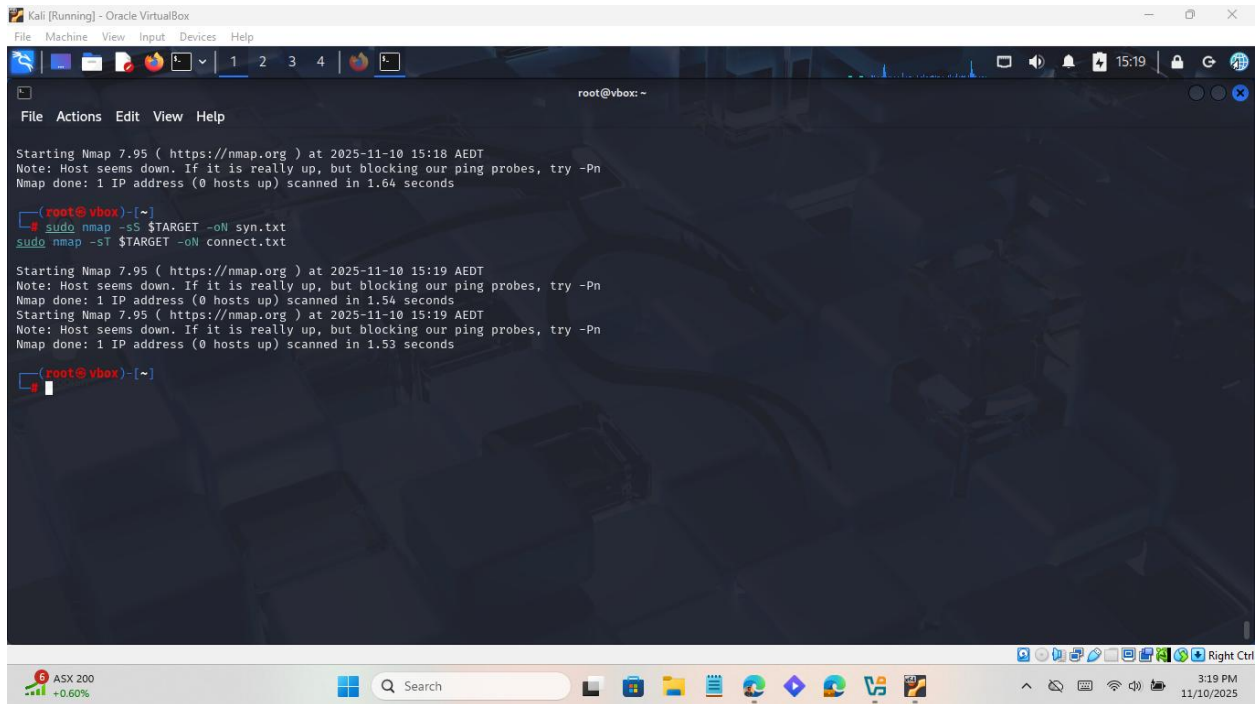
I chose one target IP from the sweep and run service scan


```
Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@vbox: ~
File Actions Edit View Help
root@vbox)~# grep "Host is up" hosts.txt | wc -l
3
root@vbox)~# TARGET=10.0.2.6
root@vbox)~# sudo nmap -sV $TARGET -oN services.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:17 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.84 seconds
root@vbox)~#
```

4.1.5. OS fingerprinting

```
Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
root@vbox: ~
File Actions Edit View Help
root@vbox)~# sudo nmap -sV $TARGET -oN services.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:17 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.84 seconds
root@vbox)~# sudo nmap -O $TARGET -oN os.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:18 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.64 seconds
root@vbox)~#
```

4.1.6. SYN stealth vs TCP connect



The screenshot shows a Kali Linux terminal window with the following commands and output:

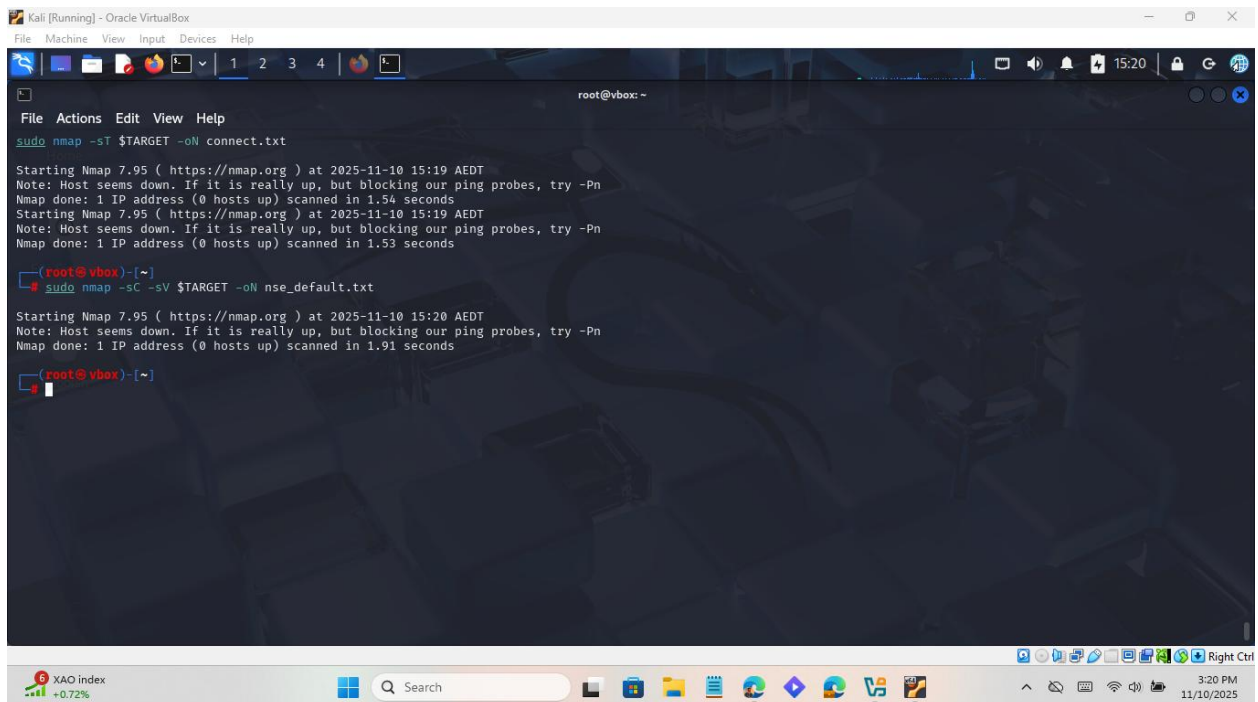
```
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:18 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.64 seconds

(root@vbox)-[~]
# sudo nmap -sS $TARGET -oN syn.txt
sudo nmap -sT $TARGET -oN connect.txt

Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:19 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.54 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:19 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.53 seconds

(root@vbox)-[~]
```

4.1.7. NSE default scripts (vulnerability indications)



The screenshot shows a Kali Linux terminal window with the following commands and output:

```
sudo nmap -sT $TARGET -oN connect.txt

Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:19 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.54 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:19 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.53 seconds

(root@vbox)-[~]
# sudo nmap -sC -sV $TARGET -oN nse_default.txt

Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-10 15:20 AEDT
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.91 seconds

(root@vbox)-[~]
```

Reflection snippet to include

Reconnaissance helps defenders identify exposed services, outdated versions, and misconfigurations before attackers exploit them. Pairing recon with patching, firewall rules, and monitoring significantly reduces the attack surface.